

# Could a transformer recover predictive state?

Belief-geometry primer · before Notebook 2 · 8–10 minutes

ILIAD Intensive · Computational Mechanics

# The discovery puzzle

**A transformer is trained only on next-token loss. Why look for anything richer than its next-token probabilities?**

History 01

Belief (0, 0, 1)

Next token (1/2, 1/2)

History 10

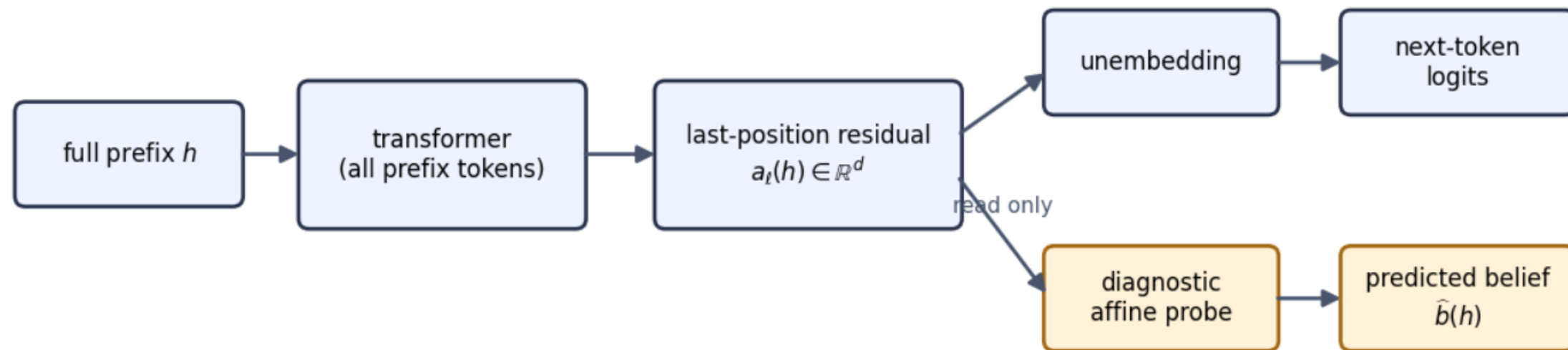
Belief (1/2, 1/2, 0)

Next token (1/2, 1/2)

**The objective scores one step at a time; good computation can still depend on distinctions that matter later.**

# Where can we look without changing the model?

## THE TEST — does a transformer encode belief geometry?



Ground truth for comparison: exact HMM belief  $b(h) = \Pr(S_{|h|} \mid h)$

**A probe is an external diagnostic fitted after training—not part of the transformer's forward pass.**

# Climb an evidence ladder, not a picture gallery

- 1 Fit:** can a simple diagnostic recover belief coordinates? starting clue
- 2 Generalize:** does it work on held-out histories? structure
- 3 Discriminate:** does it beat simpler next-token or covariate explanations? specificity
- 4 Intervene:** does changing the decoded direction change computation as predicted? causal use

# Predict the controls before seeing results

## Held-out histories

Does one map extrapolate?

## Shuffled pairing

Could high dimension fake the fit?

## Training checkpoints

Does structure emerge with learning?

## Simple baselines

Would token, length, or next-token probabilities suffice?

### PROVENANCE RULE

Keep **teaching simulation**, **reported paper result**, and **our inference** visibly separate.

# Your route: question first, reveal last

hypothesis



paired data



probe



controls



matched-prediction test

## OPEN NOTEBOOK 2

Your job is to design the test and predict its failure modes before the reported result is shown.